

Department of Computing and Mathematics Computing and Digital Technology Postgraduate Programmes Terms of Reference Coversheet	
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Signature and date student:	JERUSHA JOHANNA JANETRIN JULIES DAVID DASS 27/05/26
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This sheet should be attached to the front of the completed ToR and uploaded with it to Moodle.

Navigating the UK Property Market

Predictive Analysis of House Price Index Trends with Affordability Modelling and Explainable Deep Learning

1. Project Title

Navigating the UK Property Market: Predictive Analysis of House Price Index Trends with Affordability Modelling and Explainable Deep Learning.

This project builds on the original brief set by the supervisor, Dr Nishanthi Abeynayake, and extends it in two directions. Rather than predicting the national House Price Index as a single number, the forecasting is done at regional level across the UK. The prediction target is also extended beyond raw price to include a housing affordability measure the ratio of regional house prices to regional median wages which is what actually matters to buyers and policymakers. The project compares traditional statistical models (ARIMA, SARIMA) with a modern deep learning approach (LSTM) and introduces a multi-task version that learns both targets simultaneously. A SHAP interpretability layer then explains which economic factors are driving each forecast.

2. Course-Specific Learning Outcomes

In line with British Computer Society requirements, the project needs to fit clearly within the degree programme. The learning outcomes below are taken from the MSc Data Science specification and each is addressed directly below.

1. *“Apply a systematic methodology to the selection, implementation, and evaluation of data science techniques for solving real-world problems.”*

This project tests four models under identical conditions: a naive baseline, ARIMA, SARIMA, and an LSTM. The multi-task extension then adds a second prediction target on top of that. Every modelling choice will be written up and justified. The end result is a fair comparison rather than just picking whichever model looked best after the fact.

2. *“Analyse, pre-process, and engineer features from complex, heterogeneous datasets.”*

The project draws data from four separate government sources: Land Registry HPI (monthly regional prices), ONS ASHE (annual regional earnings), ONS macroeconomic series (CPI, unemployment, mortgage approvals), and the Bank of England base rate. These come in different formats and at different time frequencies, so turning them into a single usable dataset requires careful alignment, stationarity testing, and lag feature engineering.

3. *“Select, justify, and apply appropriate machine learning and statistical methods, demonstrating awareness of their assumptions and limitations.”*

ARIMA and SARIMA work well when the data behaves linearly, but UK house prices clearly do not they jump during the 2008 crash, COVID-19, and the 2022 interest rate rises. LSTM handles this better because it makes no linearity assumption. The multi-task version goes one step further by training a single model to explain both price and affordability at once, which forces it to learn their shared economic relationship rather than treating them as unrelated.

4. *“Critically evaluate model performance and communicate findings to technical and non-technical audiences.”*

The project uses standard forecasting metrics (RMSE, MAE, MAPE) and a Diebold-Mariano statistical test to check whether any improvements are genuine rather than lucky. The SHAP plots will turn the model’s internal reasoning into charts that show, for example, whether rising interest rates are pushing down affordability in the way economics predicts. That is useful both for someone reading a technical dissertation and for a policymaker trying to understand housing trends.

3. Project Background

What the literature does and why it is not enough.

Most published work on UK house price prediction focuses on the national HPI number and uses either ARIMA-type statistical models or, more recently, LSTM networks. The papers by Fan et al. (2018) and similar studies show that LSTM generally outperforms ARIMA on housing data, which makes sense because house prices are not a smooth, predictable series they respond to economic shocks in ways a linear model cannot capture.

However, after reading through the existing literature, three things stand out that nobody has done together in a single study.

The first is regional prediction. Nearly every paper predicts a single national figure. But a flat in London and a terraced house in Blackburn are not part of the same market. They respond to different economic pressures and serve different populations. Predicting a regional index tells a more honest story.

The second is affordability as the target. Every paper reviewed predicts raw price or price growth. That tells you whether homes are getting more expensive, but not whether ordinary workers can actually buy them. If prices rise by 5% in Manchester but wages also rise by 5%, affordability has not changed. If wages stay flat, it has gotten worse. The ratio of price to local median wage available from the Land Registry and ONS ASHE respectively captures this properly. No ML study of UK housing uses regional wages as a feature or affordability as a prediction target.

The third is multi-task learning. HPI and the affordability ratio are not independent: they share the same economic drivers (interest rates, employment, housing supply). A model that has to explain both simultaneously should, in theory, learn that shared structure better than a model trained on one target alone. This is a well-established idea in machine learning (Caruana, 1997) but has not been applied to UK housing.

Why this study fills a gap. Looking at the literature, no published paper was found that (a) forecasts UK housing affordability at regional level, (b) uses ONS regional wage data as a feature, and (c) applies a multi-task deep learning approach to the joint HPI and affordability prediction problem. The data needed to do this is freely available from UK Government sources. That is why this is the right project for an MSc Data Science dissertation.

Why the data is straightforward to obtain.

The Land Registry publishes the full UK HPI dataset, broken down by region, as a free CSV download going back to 1995. ONS publishes ASHE earnings data by region every year, also free. The Bank of England base rate and ONS macroeconomic series are downloadable from their websites without registration. None of this requires purchasing a dataset, applying for access, or signing a data sharing agreement.

4. Aim

The aim of this project is to forecast UK regional housing affordability measured as the ratio of regional HPI to regional median annual wages using a multi-task LSTM model that also predicts regional HPI, and to compare this against ARIMA, SARIMA, and single-task LSTM baselines in a rigorous, reproducible evaluation. A SHAP interpretability analysis will then identify which economic and wage-related features drive the model's predictions in different regions and time periods.

The central research question is:

Does a multi-task LSTM that jointly predicts regional UK House Price Index and a regional housing affordability ratio trained with ONS wage growth as an additional feature produce more accurate and more interpretable forecasts than standalone ARIMA, SARIMA, and single-task LSTM models across UK regions, and which macroeconomic features does SHAP identify as the main drivers of regional affordability change?

A secondary question concerns whether the economic drivers of affordability differ between London and northern regions. London has very high prices relative to wages already. Northern regions have lower wages but are seeing faster price growth. If the SHAP results reveal different driver patterns across regions, that would be a meaningful contribution to housing policy research.

5. Objectives

1. **Collect the data and build the affordability ratio.** The project downloads the Land Registry regional monthly HPI series (1995–2025) and ONS ASHE annual regional median

earnings. Regional HPI is divided by interpolated regional earnings to produce a monthly affordability ratio for each of nine English regions and Wales. All download dates, sources, and transformation steps are recorded in a reproducible notebook.

2. **Bring in macroeconomic features and prepare them for modelling.** ONS CPI, unemployment, and mortgage approval series, and the Bank of England base rate are downloaded and aligned to monthly frequency. Stationarity tests (ADF and KPSS) are applied to all series. Lag features are created at 1, 3, 6, and 12 months, along with rolling averages and standard deviations over 3-month and 12-month windows.
3. **Build and evaluate the baseline models.** Four baselines are implemented: (a) naive model using last observed value as the forecast; (b) ARIMA fitted using Box-Jenkins identification; (c) SARIMA with seasonal order selected by grid search; (d) single-task LSTM predicting HPI only. All four are run under the same train/validation/test split and are evaluated with the same metrics.
4. **Build and evaluate the multi-task LSTM.** The multi-task model is an LSTM with a shared encoder and two output heads: one for HPI, one for the affordability ratio. It is trained jointly with a weighted loss: $\mathcal{L} = \alpha \cdot \mathcal{L}_{\text{HPI}} + (1 - \alpha) \cdot \mathcal{L}_{\text{afford}}$, where α is tuned on the validation set. This is compared directly against the single-task LSTM to show what the joint training is actually adding.
5. **Run the full evaluation across all regions.** RMSE, MAE, and MAPE are computed for both prediction targets. The Diebold-Mariano test checks whether differences between models are statistically significant. A rolling one-step-ahead backtest over the last 24 months of data, to see whether results hold up over time or only look good on the test split.
6. **Explain the best model using SHAP.** SHAP DeepExplainer produces feature importance rankings and per-prediction attribution values. Two specific checks are made: (a) do wage growth features rank higher in regions where affordability has worsened most; (b) does the base rate feature attribution increase during 2022–2024, when the Bank of England was raising rates rapidly. If both are true, the model has learned genuine economic relationships, not just patterns that happen to fit the training data.
7. **Make everything reproducible.** A public GitHub repository is maintained with regular commits. The software environment is pinned in a `requirements.txt`. A single Jupyter notebook downloads the data, trains the models, and produces all evaluation tables and SHAP charts from scratch.

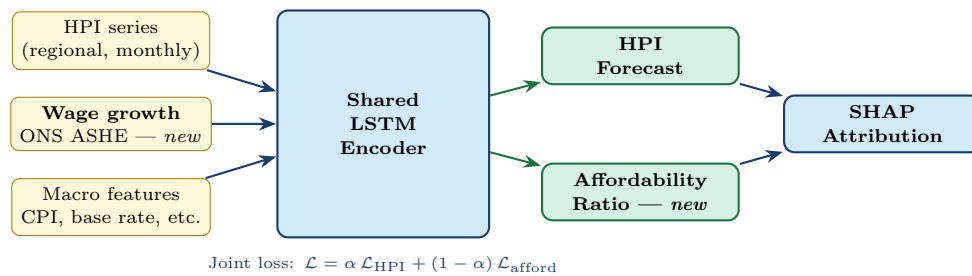


Figure 1: The multi-task LSTM architecture. One shared encoder reads all the input features; two output heads produce the HPI forecast and the affordability ratio forecast simultaneously. The gold input boxes are the novel elements not used in any existing UK housing ML study.

6. Evaluation Plan

A fixed train/validation/test split is used to keep the comparison fair across all models. Training runs from January 1995 to December 2022 (around 336 months per region). The validation set covers January 2023 to June 2024 and is used only for tuning hyperparameters. The test set covers July to December 2024 and is not touched until the very end.

Table 1: Evaluation metrics used in this project

Metric	What it measures	Rationale
RMSE	Average size of prediction errors, with larger errors weighted more	Standard metric in the forecasting literature; directly comparable to published papers
MAE	Average size of prediction errors, all treated equally	Easier to interpret than RMSE; gives the typical error in index points
MAPE	Errors as a percentage of the actual values	Makes comparisons fair across regions where the HPI level differs significantly
DM test	Statistical test for whether one model's errors are genuinely smaller than another's	Without this, a small improvement in RMSE could just be random variation rather than a real difference

A rolling one-step-ahead backtest is also conducted over the last 24 months of data. This means re-fitting the model month by month and recording each forecast. It is a more realistic test than a single train-test split because it shows whether the model stays reliable over time.

Success criteria. The project is judged successful if: (a) the multi-task LSTM beats the best single-task baseline on the affordability ratio target in at least 6 of 10 regions, confirmed by the Diebold-Mariano test at $p < 0.05$; (b) the SHAP results make economic sense (wage features matter more where wages are lowest; base rate attribution rises during 2022–2024); (c) the full pipeline runs end-to-end from the public GitHub repository without modification. If the multi-task model turns out not to be better, this is reported honestly in the dissertation that result is still useful because it tells future researchers not to bother with the extra complexity for this particular problem.

7. Anticipated Problems

- **The wage data is annual but everything else is monthly.** ONS ASHE earnings are published once a year. The series is interpolated to monthly frequency using linear interpolation as the primary approach; a sensitivity analysis then checks whether forward-fill or cubic spline interpolation makes a meaningful difference. If the interpolation method matters a lot, this is documented as a limitation.
- **The affordability ratio might amplify measurement error.** Dividing two imperfect series can amplify measurement error. Bootstrap confidence intervals are produced for ratio forecasts, and the limitation is discussed explicitly in the dissertation.
- **Balancing the two loss functions in the multi-task model.** If the HPI loss is weighted too heavily, the model will not learn affordability well, and vice versa. The parameter α is treated as a hyperparameter and is searched over the range 0.3 to 0.7 in steps of 0.1 on the validation set.
- **Overfitting with not much training data.** Monthly data from 1995 gives roughly 336 observations per region. That is not a lot for a deep learning model. Dropout regularisation, early stopping, and L2 weight regularisation are applied. If per-region training still overfits, a cross-region pooled training approach is tested.
- **SHAP is slow on LSTM models.** Computing SHAP values for sequence models is slow. A background reference set of 200 sampled time points is used rather than the full training set; the approximation error is documented.
- **Scope could get too big.** Ten regions and five models is an ambitious evaluation matrix. If time pressure arises, the scope contracts to five regions London, North West, Yorkshire, South East, and Wales without changing the methodology.
- **Details still being finalised with the supervisor.** The exact look-back window length for the LSTM, and whether to include a transformer-based comparison model, are under discussion with Dr Abeynayake. These details will be confirmed during the first two weeks of the project.

8. Required Resources

All resources required are free and publicly available.

Table 2: Resources required for this project

Category	Detail
Data	Land Registry UK HPI (regional monthly, 1995–2025, Open Government Licence, free CSV). ONS ASHE regional earnings (annual, free). ONS CPI, unemployment, and mortgage approval series (free). Bank of England base rate (free).
Software	Python 3.11 with <code>pandas</code> , <code>numpy</code> , <code>statsmodels</code> for ARIMA and SARIMA, <code>tensorflow/keras</code> for the LSTM models, <code>shap</code> for interpretability, <code>matplotlib</code> and <code>scipy</code> for analysis and visualisation. All open-source. Environment recorded in <code>requirements.txt</code> .
Computation	Standard personal computer for data processing and ARIMA models. Google Colab (free GPU tier) for LSTM training where needed. No paid cloud computing or HPC access is required.
Version control	Public GitHub repository with regular commits; included as a compulsory appendix in the dissertation.
Writing	The dissertation is written in $\text{\LaTeX}$.
Literature	MMU Library portal, Google Scholar, IEEE Xplore, ACM Digital Library, and SSRN for economics working papers.

9. Schedule

Table 3: Project timetable

Period	Milestones
Weeks 1–2 (by 5 Jun)	ToR submitted (29 May). EthOS application submitted. All four datasets downloaded and checked for gaps. Affordability ratio constructed for all regions. Stationarity tests completed. GitHub repository and $\text{\LaTeX}$ skeleton established.
Weeks 3–4 (Jun)	Ethics approval confirmed. Literature review: time-series forecasting, LSTM, multi-task learning, and UK housing ML. Background chapter first draft completed.
Weeks 5–7 (Jun–Jul)	ARIMA and SARIMA baselines built and validated across all regions. Feature engineering finalised. Single-task LSTM built and tuned on validation set. Methodology chapter drafted.
Weeks 8–10 (Jul–Aug)	Multi-task LSTM built. Loss weight tuned. Full evaluation run: RMSE, MAE, MAPE, Diebold-Mariano tests, and rolling backtest. SHAP values computed. Results chapter drafted.
Weeks 11–13 (Aug)	Introduction and Conclusions written. Full draft submitted to supervisor by approximately 18 August. Appendices assembled.
Weeks 14–16 (Aug–Sep)	Supervisor feedback incorporated. References checked. Proofreading completed. Dissertation submitted by 11 September 2026 .
Week 17 (Sep)	Presentation prepared and delivered on 18 September 2026.

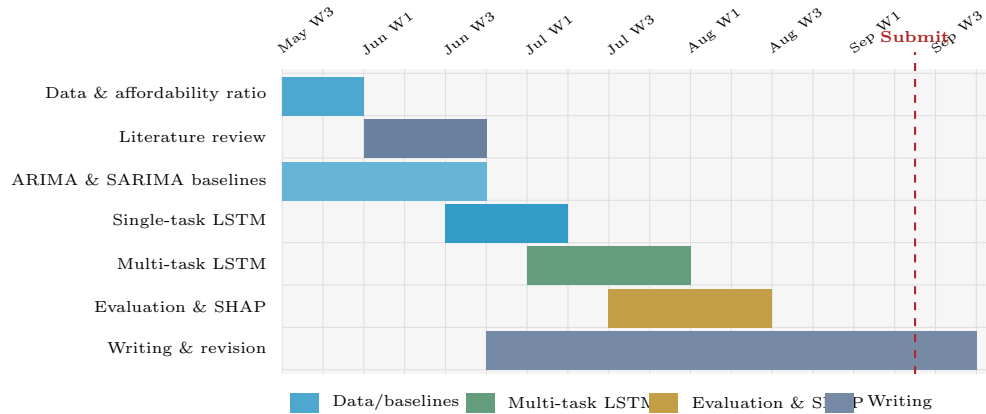


Figure 2: Project timetable. The green bar shows the novel multi-task LSTM phase. The red dashed line is the submission deadline of 11 September 2026.

10. Ethics

The project uses only publicly available data published by UK government bodies. None of it contains personal or identifiable information. Specifically:

- Land Registry UK HPI — Open Government Licence v3.0, freely downloadable, aggregated price statistics only, no individual transaction records.
- ONS ASHE regional earnings — Open Government Licence, regional summary statistics, no individual-level salary data.
- ONS macroeconomic series (CPI, unemployment, mortgage approvals) — Open Government Licence, aggregate national statistics.
- Bank of England base rate — public domain, freely available from the BoE Statistical Interactive Database.

There are no human participants in this project. No data is collected from people, conducting interviews or surveys, or handling any sensitive personal information. The project is entirely computational and uses secondary data that is already in the public domain.

This means the project falls under the EthOS proportional self-assessment route at Manchester Metropolitan University. An application has been submitted through the EthOS system, with approval anticipated by 5 June 2026. The approval number will go into the originality declaration of the dissertation, and the full approval record will be included as a compulsory appendix alongside the GitHub repository link and a copy of this Terms of Reference document.

Ethics approval number: *To be confirmed — anticipated by 5 June 2026.*

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